Counts and averages: GROUP BY



Data engineering · Level 3 · Lesson 3

Today we will

- Summarise with COUNT, AVG, MIN and MAX.
- Make one row per group with GROUP BY, and filter groups with HAVING.
- Handle NULL, and check summaries for glitches.

Remember

- GROUP BY: one row per group.
- WHERE before grouping, HAVING after.
- Check MIN and MAX before trusting AVG.

Grouping by hand

1. Warm-up (5 min): How long would this take by hand?
2. Teach (8 min): Summaries, GROUP BY and HAVING.
3. Group (12 min): By birth month.
4. Filter (10 min): HAVING on the board.
5. Wrap-up (5 min): The same in SQL.

Pupils may choose a month instead of giving their real birth month. Keep groups small enough to count quickly, and have one pupil whose travel time is “not sure” to show what NULL does.

Let’s discuss

1. Why can’t we select each pupil’s name in a GROUP BY class query?
2. What’s the difference between COUNT(*) and COUNT(mark)?
3. Why look at MIN and MAX before an average?

Exit ticket

- Write a query for the number of pupils in each class, keeping only classes with more than 30.